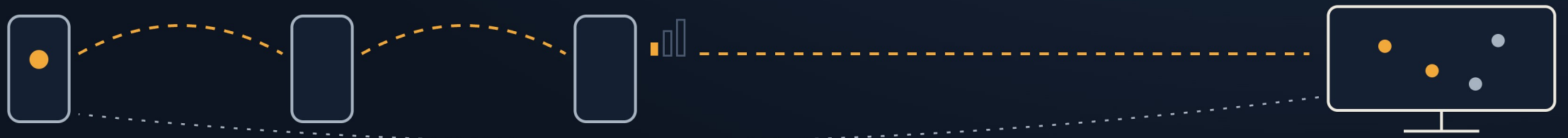
GOOGLE CLOUD AI BUILDER CUP 2026 · SUSTAINABILITY & SOCIAL IMPACT

The tower fell at 21:40. The needs didn't.

Relay carries a cry for help across phones with no signal, and brings the answer back.



Relay · Assembly Lines
Harshvardhan Singh
Manas Choudhary

relay.harshvs.com

01 · THE PROBLEM

48 hours. 6 boats. Zero signal.

After landfall, help exists. It just can’t find the people who need it.

THE USER · PURI, ODISHA

No bars on a mid-tier device. Her mother needs insulin, somewhere near the school road.

THE DUTY OFFICER · DISTRICT CONTROL ROOM

A map, forty volunteers and six boats. No idea which village to send them to first.

~7,000 of 14,167

cell tower sites in West Bengal went dark after Cyclone Amphan, 2020

135

Philippine cities and towns lost telecoms after Typhoon Odette, 2021

VHF radio

is what Puri’s officials fell back on after Cyclone Fani, 2019, when mobiles failed within hours

Sources: The Quint, citing TAIPA (May 2020) · Philstar Interaksyon (Dec 2021) · Hindustan Times (13 May 2019)

02 · WHY TODAY'S CHANNELS FAIL

Every lifeline assumes the tower is standing.

Calls and SMS need a tower; mesh chat needs a contact. Relay needs neither.

Channel	Works with towers down	Reaches the control room	Arrives ready to act on	Confirms help is coming
Calls and SMS	No	Yes	No	Sometimes
Emergency helplines	No	Yes	No	No
Satellite phones	Officials only	Yes	No	No
Mesh chat apps	Yes	No: contacts only	No: free text	No
Relay	Yes	Yes, via gateway	Yes: a ticket	Yes, hops back

During Cyclone Fani, even satellite phones in Puri struggled; officials fell back on police VHF radio. Source: Hindustan Times, 13 May 2019

03 · THE SOLUTION

Phone to phone to control room. And back.

A structured need hops until one phone finds signal; the confirmation rides the same path home.

01
User phone
No signal. A voice note or a tap; Gemma asks 3 questions.

→

02
Relay phones
BLE mesh, TTL 7. Store and forward; nobody opens an app.

→

03
Gateway phone
One bar of signal. Uploads tickets, carries replies.

→

04
Relay cloud
Cloud Run and ADK agents. Triage, dispatch, sitrep, translate.

→

05
Control room
Live map and queue. The officer approves in one click.

← BACK THE SAME HOPS “Boat leaving the school at 15:40. Keep her warm.”

04 · THREE MODES

Installed before the storm. Needed during it. Trusted after.

Relay has three modes, and the outage is only the middle one.

BEFORE LANDFALL

Ready

The only requirement is having the app. Users share it phone to phone over Android Nearby Share, so it spreads without a store or a signal.

`Android Nearby Share`

FIRST 48 HOURS

Relay

A voice note becomes a ticket on the phone, hops across the mesh, is triaged in the cloud, approved by a human and confirmed back.

`Gemma · ADK agents · Gemini Flash`

AFTER THE TOWERS RETURN

Recover

Every ticket and dispatch becomes evidence: a damage map by village, households reached, and a log for the state and insurers.

`Gemini Pro, multimodal`

Ready mode is the distribution plan: it’s why Relay is already on the phone when the tower goes.

05 · ARCHITECTURE

Five Google tools. Zero passengers.

Every component has a job; none is there to tick a box.

ON THE PHONE · OFFLINE

Gemma 4 E2B on LiteRT-LM/Gemma 3 270M (fallback)

Voice note → 100-byte ticket; asks the dispatcher's questions

Nearby Connections BLE mesh

Store and forward, TTL 7, ACKs

→

GATEWAY

Any phone with one bar

Syncs tickets up and replies down over HTTPS

→

RELAY CLOUD

ADK agents on Cloud Run

Triage, dispatch, sitrep

Gemini Flash + Maps grounding

Merge, geocode, explain

Gemini Pro

Sitreps and the recovery report

→

CONTROL ROOM

Adminstrator Node

Live map, priority queue, one-click approve

06 · GEN AI AT THE EDGE

60 seconds of panic. 100 bytes of signal.

Gemma turns a voice note into a ticket, offline, and asks what a dispatcher would.

Voice note, Odia, 0:58: “Ma needs insulin. We’re near the school.”

Gemma: Is she conscious?

Yes, but weak.

Gemma: How many people are with you?

Four.

```
TICKET · ≈100 BYTES
category MEDICAL
item insulin
people 4
urgency HIGH
landmark near the school
language Odia
answers conscious: yes
```

Why on the phone

There is no cloud in the first hours.

Why 100 bytes

Small enough to survive three Bluetooth hops.

If the model is late

A rule-based parser ships the same ticket.

Speed on the demo phone

Voice note: ≈5.7 s (p50) per intake turn on Samsung S24FE, Gemma 4 E2B. Typed text: ≈4.8 s (p50), same phone.

07 · GEN AI IN THE CONTROL ROOM

40 cries. 6 boats. 1 click.

Gemini agents merge, locate, prioritise and draft; the duty officer approves.

40 tickets on one cyclone night → **6** dispatches, one per team, plus an hourly sitrep → **1** human click to approve each one

Embeddings	Maps	Flash	ADK agent	Flash	Gemini Pro
Merge	**Locate**	**Prioritise**	**Allocate**	**Draft**	**Report**
Three neighbours, one family: one ticket, three reports.	“Pipili bus stand” becomes a dot on the map.	A score, with a reason a human can read.	Teams within capacity and travel time.	Orders in English and the local language.	The hourly sitrep, in the district’s format.

Demo runs on a synthetic forty-ticket cyclone night, labelled as such. Agents draft; the approve button stays human.

08 · WHERE AI EARNS ITS PLACE

Where the AI helps in the process

We mapped every stage. The radio and the return path get no AI, by design.

Ready · Flash + TTS
Warning → a plan, read aloud

Intake · Rules · Gemma 4 E2B (spike)
Speech → ticket, plus follow-ups

Mesh · No AI, by design
The radio is the constraint

Triage · Rules · Flash + Maps
Merge, geocode, explain priority

Dispatch · ADK agent
Allocate teams, draft orders

Sitrep · Gemini Pro
The report the district asks for

Reclassify · Gemini Flash
Five languages, cloud checked

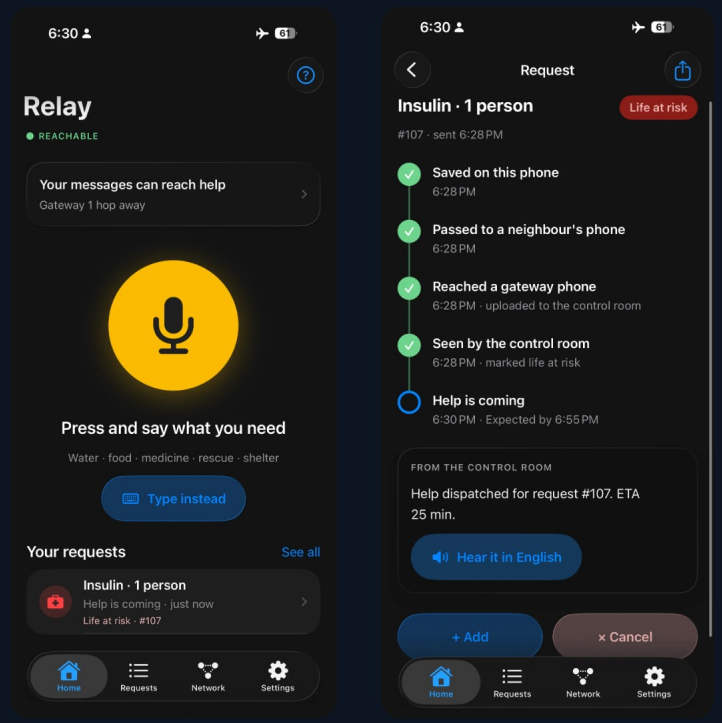
Return path · No AI, by design
A template id; the phone renders it

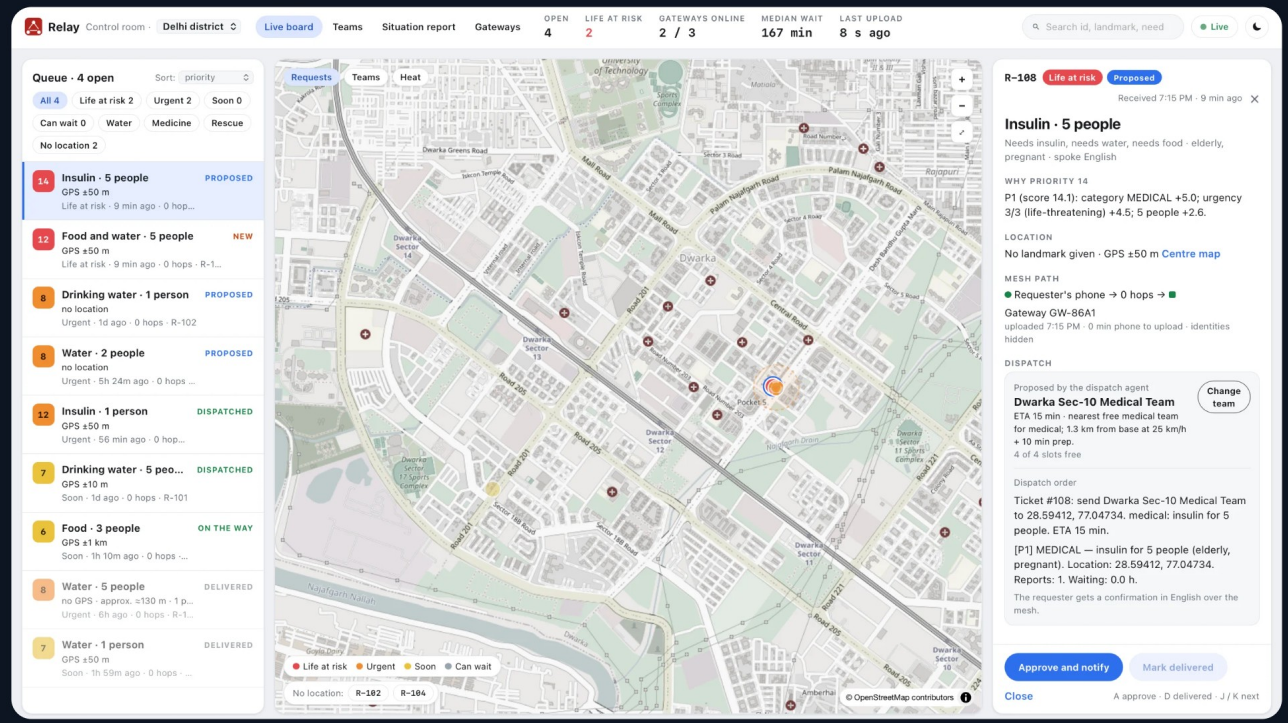
Recover · Gemini Pro
Relief report and evidence packs

09 · EXPERIENCE

One button. One queue.

No literacy, no network, no app to find for the user. A reason beside every priority for the officer.





User: one button and a voice, covering multiple Asian and Indic languages, across dialects. Status changes only on a real ACK.

Duty officer: map, queue and a reason beside every priority. One click to approve.

10 · PROTOTYPE

Not a mockup. Three phones in airplane mode.

The mesh, gateway and control room run today. Here’s what we measured.

RUNNING TODAY

- BLE mesh, store-and-forward with ACKs
- Gateway sync and localised confirmations
- Control room: map, queue, reasons, approve

ADDED FOR THE CUP

- Gemma intake conversation, offline
- Control-room agents: merge, dispatch, sitrep

SHOWN ON RECORDED DATA, LABELLED

- Photo tickets that travel as tags
- Ready mode household plans

≈100 B
per ticket on the wire

1–3
hops, phone to gateway

≈2.5 s
typed request to control room

≈5.7 s
Gemma intake turn (p50)

85%
category accuracy, English + Hindi

40 → 35
duplicate reports merged

Live control room: relay.harshvs.com · Video: Google Drive · Code: github.com/harshvs99/ai-builder-cup

11 · IMPACT

We count boats that arrived, not messages sent.

Impact is time to help and needs closed, across Asia’s cyclone belt.

119M people exposed to tropical cyclones each year, on average

85%+ of expected annual cyclone deaths fall in Bangladesh and India

5 of 5 most-exposed populations are in Asia, India and the Philippines among them

WHAT A DISTRICT PILOT MEASURES

- Minutes from ask to dispatch
- Share of tickets with a usable location
- Duplicate reports merged before dispatch
- Needs confirmed closed, by village
- Households reached, in the recovery report

Source: PreventionWeb (UNDRR), “Asia, most at risk for natural hazards, most vulnerable to cyclones”

Relay never says “help is coming” until it is.

Rules written into the app, and the risks we’ve already hit.

RULES IN THE APP

Confirm, never assume. Status changes only on a backend ACK or a human approval.

Ask about the situation. “Is she conscious?”, never a diagnosis.

Location is opt-in. The landmark is the fallback rescuers rely on.

Minimum data. No names, no numbers, no pixels; photos travel as tags.

Humans dispatch. Agents draft; the approve button stays human.

The district owns the data. Demo data is synthetic and clearable.

RISKS WE’VE HIT, AND THE FIX

BLE links flap. A rehearsed two-hop path on the exact demo phones.

Latency on mid-tier phones. Gemma 3 270M for intake; rule-based parser as fallback.

Hindi and Odia misparsed. On-phone intake, cloud re-classification, original kept for audit.

Duplicate or spoofed tickets. Message ids and merging with report counts; per-device signing next.

13 · SCALE

One district this cyclone season. The JAPAC coast next.

Runs on phones people already own; the cloud works only when there is work.

NOW · PILOT

One Odisha district

- District control room on Relay
- Ready mode before the warning
- Baseline vs Relay: time to help

NEXT · STATE

Every coastal district

- SMS and IVR for feature phones
- Per-device signing
- Multi-district control rooms

THEN · JAPAC

The cyclone belt

- Bangladesh, Philippines, Vietnam
- Data-mule mode between islands
- Rumour checks before broadcast

Cloud cost per event: < ₹1,000 (est.)

Per district, per season: < ₹2,500 (est.)

New hardware: none

RELAY · SUSTAINABILITY & SOCIAL IMPACT

The tower might fall again.
But this time, the help will arrive.



Assembly Lines — Harshvardhan Singh, Manas Choudhary

Video: Google Drive
Code: github.com/harshvs99/ai-builder-cup